OCP SPECIAL FOCUS:
PHOTONICS

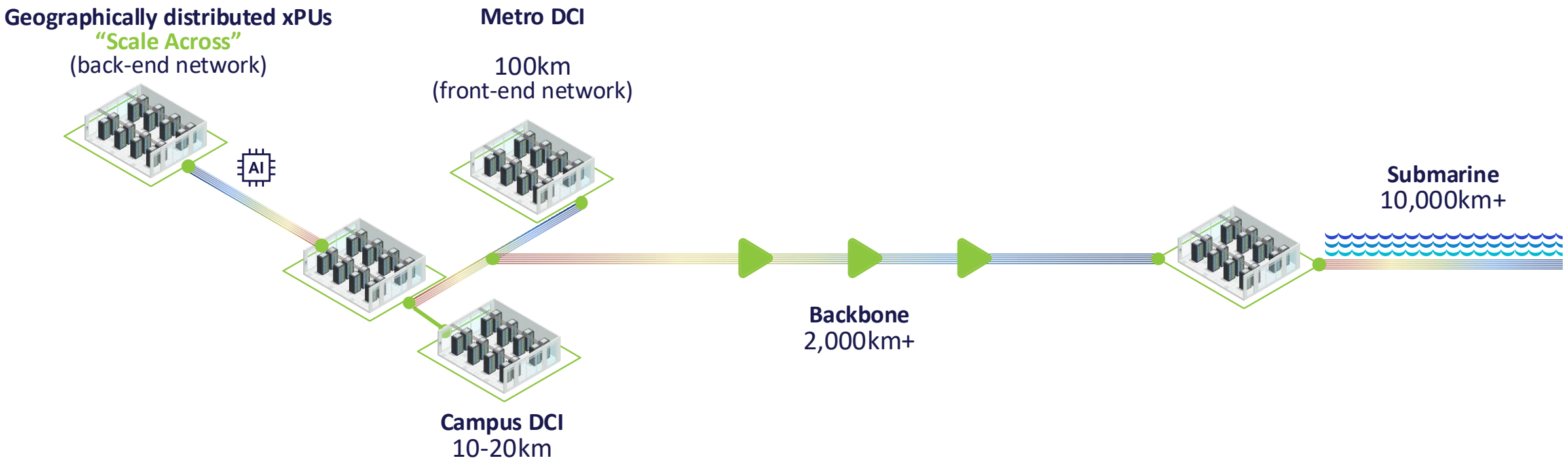


# The Future of Coherent Optics in the AI Era

Frank Chang *Sr Director, Interconnects PLM, Ciena*

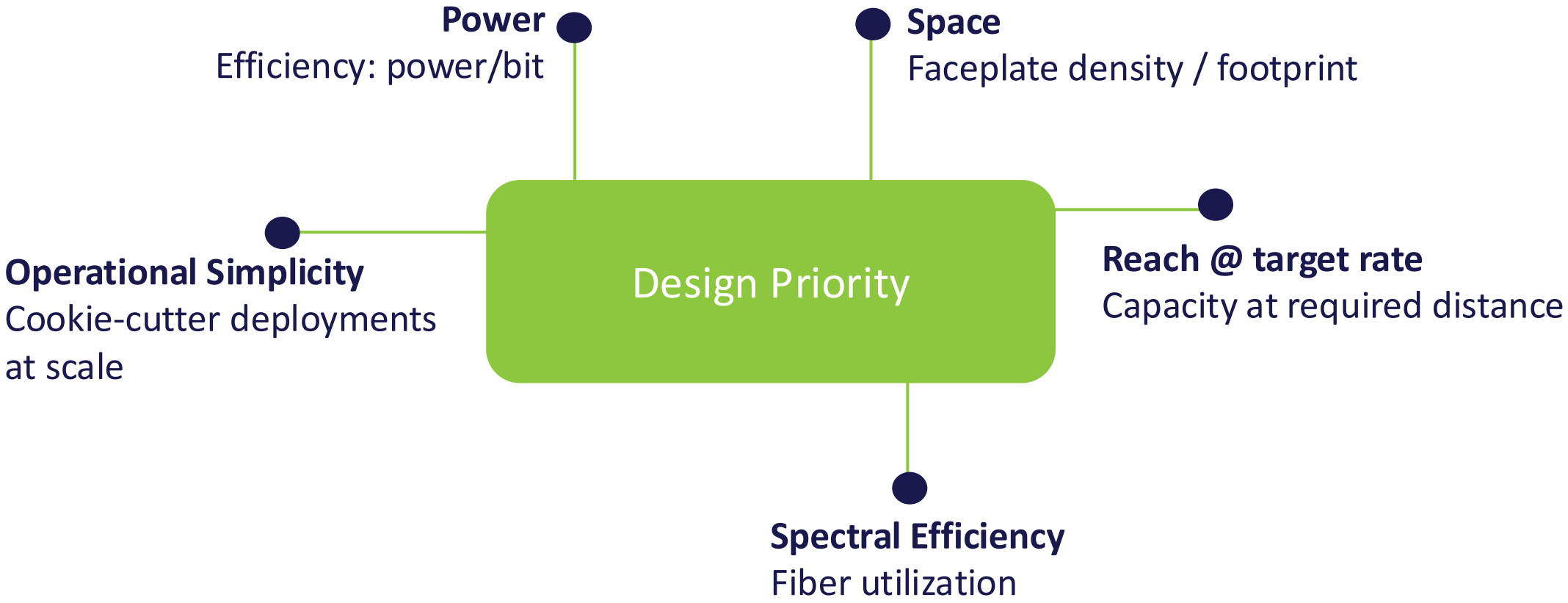


# The expanding DCI landscape requires more than one optical architecture




Different DCI applications are creating fundamentally different optical requirements.

# The right coherent architecture depends on what matters most



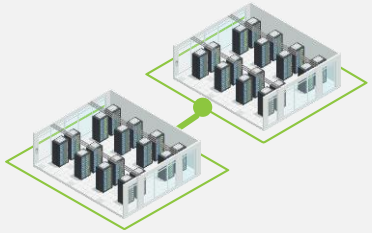
# Coherent solutions increasingly overlap across DCI applications





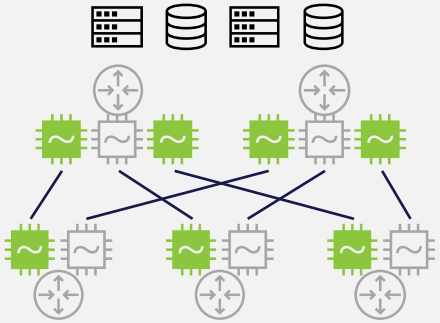
| Around the DC | Metro/Regional | Core |
|---|---|---|

### Campus DCI

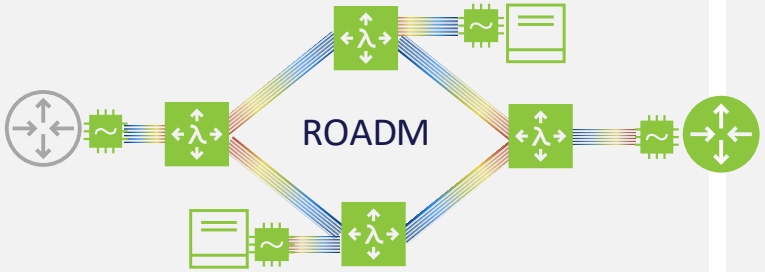


2-20km

### DCI



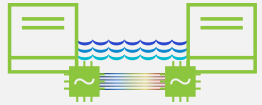
### Multi-span ROADM




### Long Haul



### Subsea



# Submarine: Different priorities drive different coherent choices








| | Coherent Pluggable Transceiver | Performance Transponder |
|---|---|---|
| Platform | WaveLogic 6 Nano | WaveLogic 6 Extreme |
| Capacity | 23.2 Tb/s | 27 Tb/s (+16%) |
| Channels | 29 x 800 Gb/s (150GHz) | 27 x 1 Tb/s (162GHz) |

Power & space optimization → Pluggables

Spectral Efficiency Optimization → Transponders

# Metro DCI: Different priorities drive different coherent choices





| | 800G Coherent Pluggable Transceiver | 1.6 Tb/s Performance Transponder |
|---|---|---|
| Technology | WaveLogic 6 Nano | WaveLogic 6 Extreme |
| Capacity | 51.2 Tb/s | 76.8 Tb/s (+50%) |
| Channels | 64 x 800 Gb/s (150GHz) | 48 x 1.6 Tb/s (200GHz) |

Power & space optimization → Pluggables

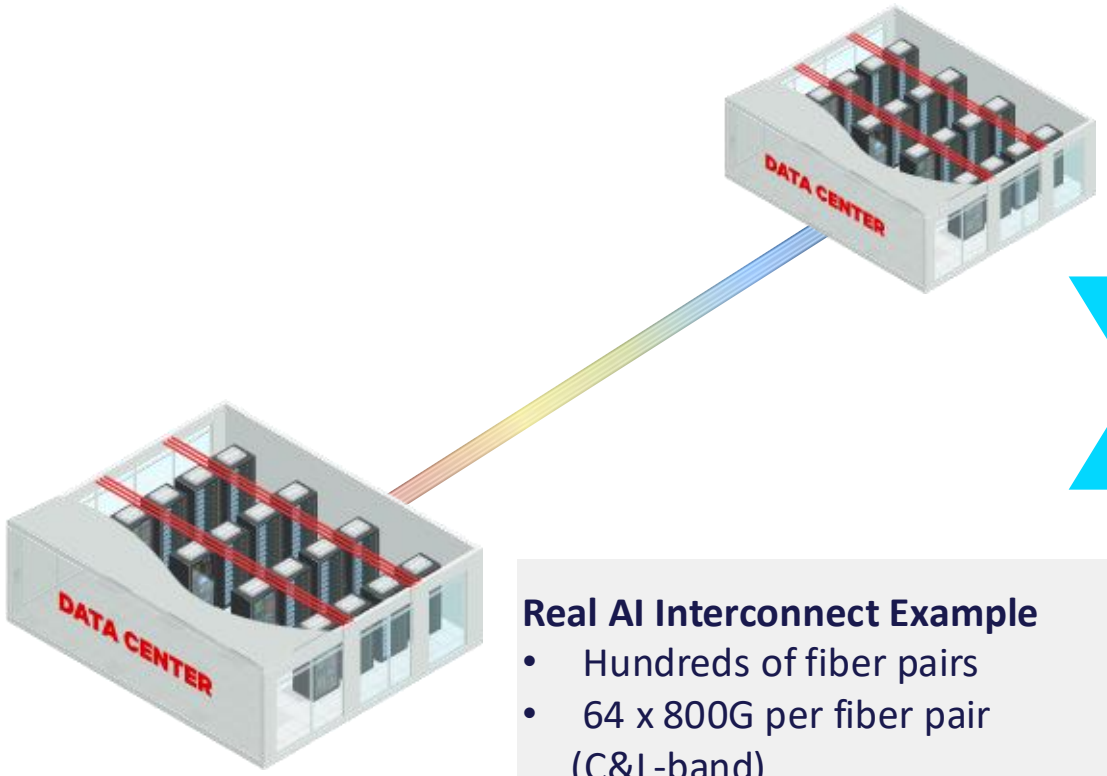
Evolution: 1600ZR, 1600ZR+, 1600G Coherent-Lite, …

Spectral Efficiency Optimization → Transponders

Evolution: Network sensing, 300/400GHz designs

# Scale-Across networking has arrived

## SCALE-ACROSS SOLUTION




**Real AI Interconnect Example**

- Hundreds of fiber pairs
- 64 x 800G per fiber pair (C&L-band)

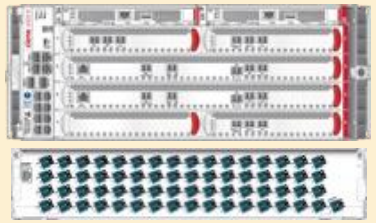
## WaveLogic coherent optics



- WaveLogic 6 Nano 800G ZR+ with interoperable PCS
- C- and L-Band variants

## RLS modern line system

RLS C&L terminal config



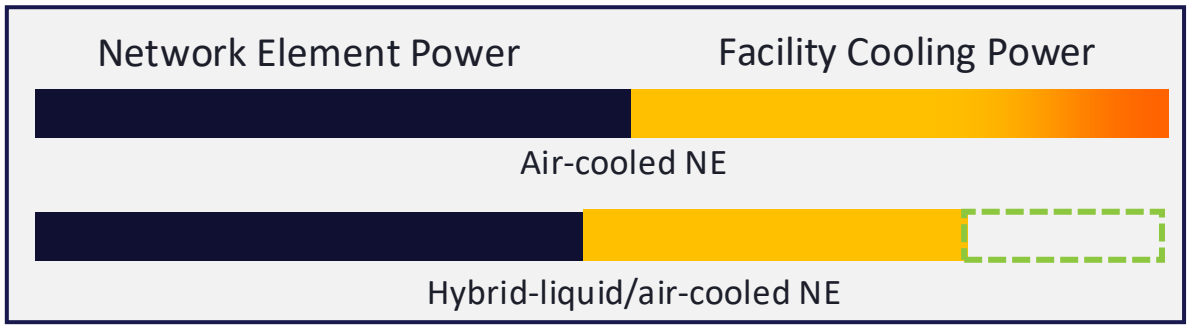
RLS Hyper-Rail

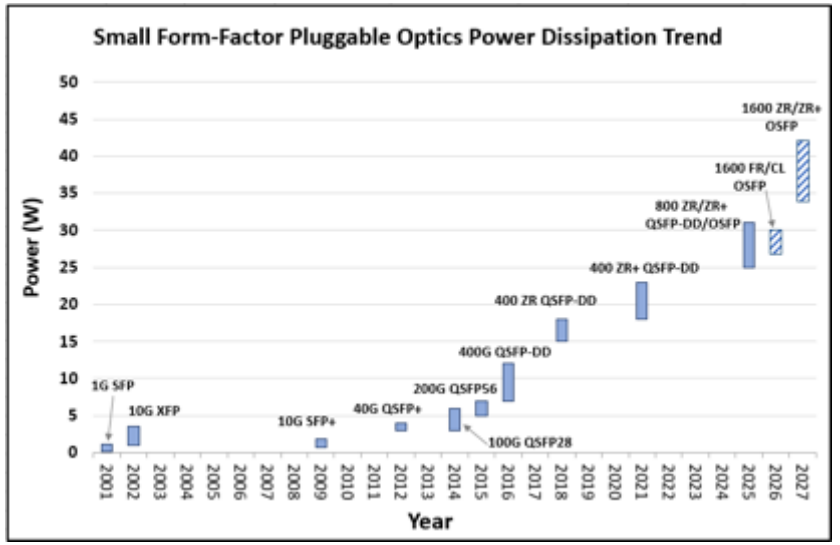


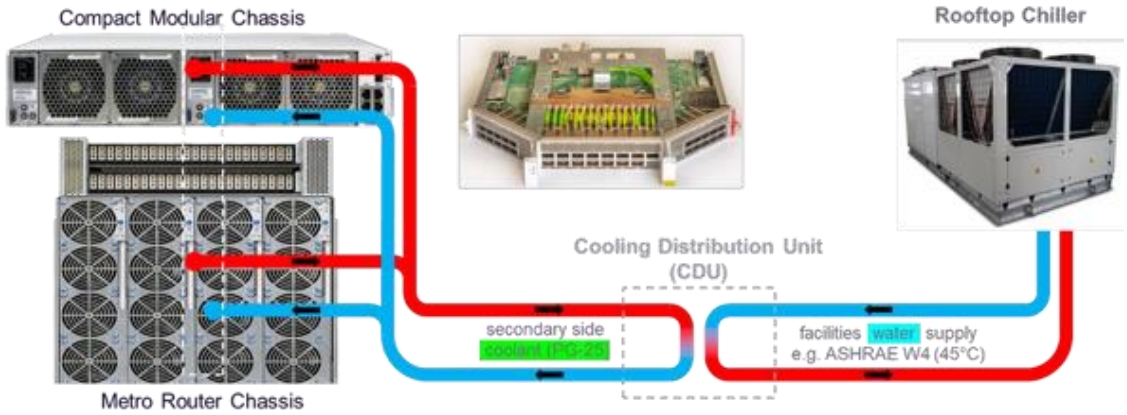
16 x C- and L-Band
fiber pairs in compact chassis

# The power wall is driving the transition to liquid cooling

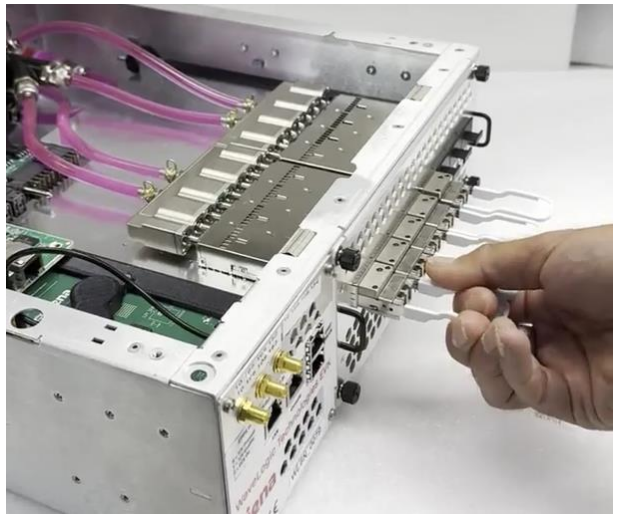
- OPEX savings thru facility energy consumption, water consumption, GHG emissions
- Eliminates fan power consumption and system fan noise.




Liquid cooling enables additional ASICs per RU

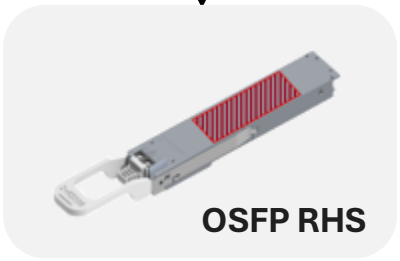




12.8T XPO

# Power and density are reshaping optical form factors

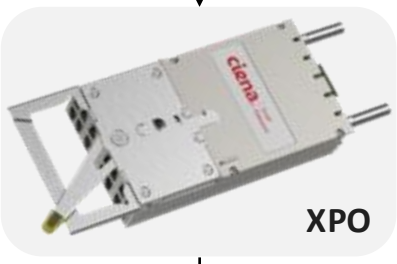


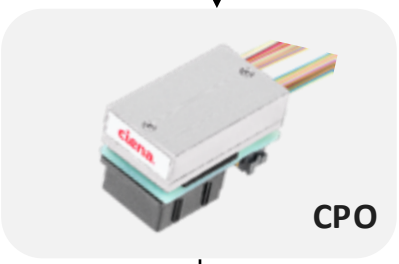

- Higher power design MSA-standard form factor
- System liquid cooling



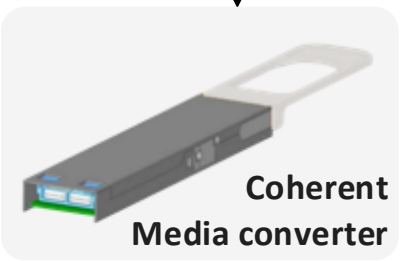

- Direct to plug liquid cooling
- Scales to higher power design




- New form factor
- 2.4 x density




- Higher I/O density




- Coherent transponder pluggable



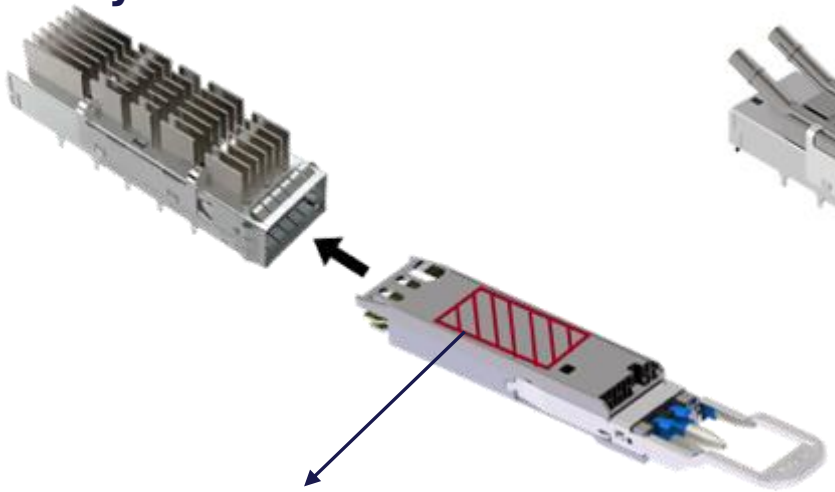

- Full-fiber deployment
- Operational simplicity

More power. More density. More integration.
More choice.

# Direct-to-plug cooling extends the power envelope for coherent pluggables

## Riding Air-Cooled Solution

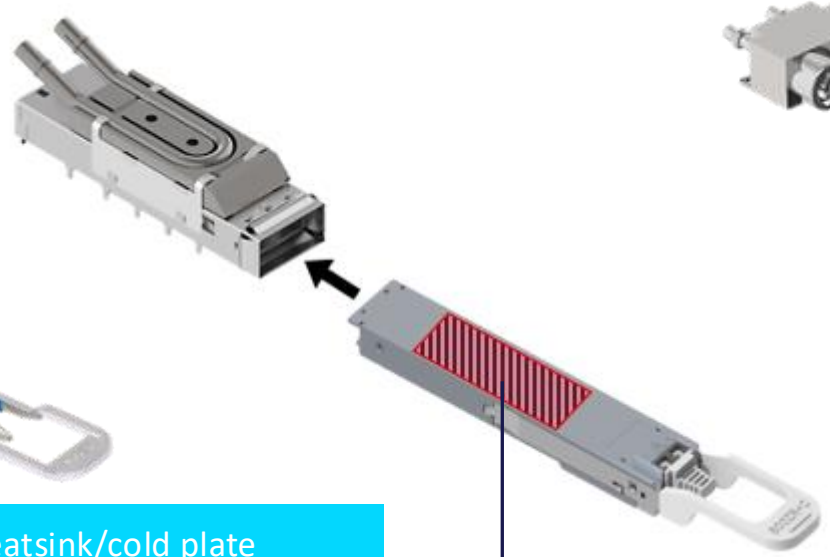
**Today's solution**

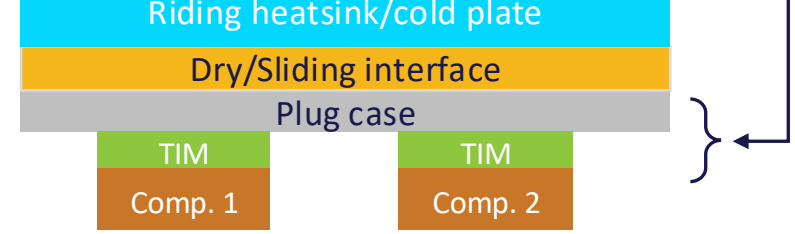


x **Thermal Penalty**

## Riding Liquid-Cooled Solution

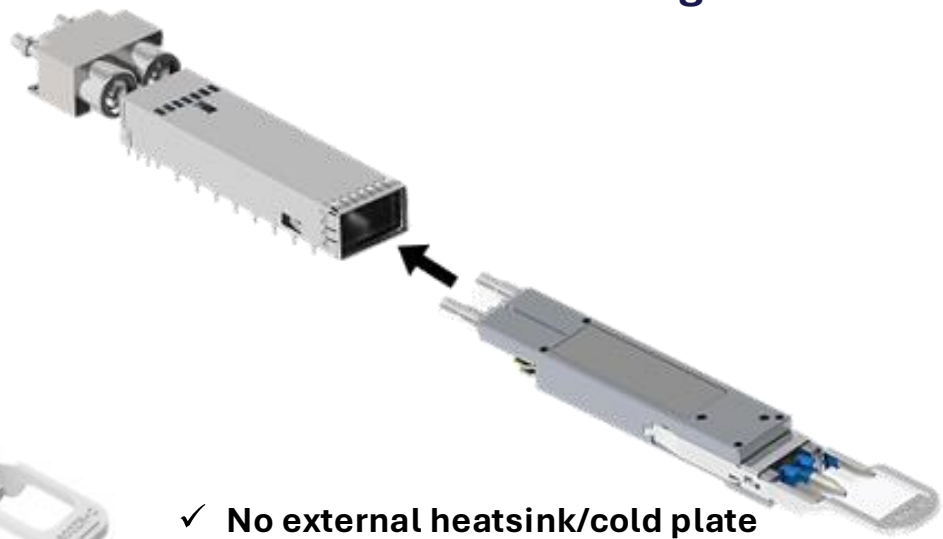
**Existing OSFP MSA-compliant**






## Direct-To-Plug Liquid-Cooled Solution

**New mechanical design**



- ✓ **No external heatsink/cold plate**
- ✓ **No dry/sliding contact**



# XPO scales density while preserving front-panel pluggability

- Scaling routers and pluggables

- Standard, flexible liquid-cooled 12.8T pluggable
- Higher density routers – 200T in 1RU
- Robust ecosystem with XPO MSA
- Flexible thermal envelope enables products for: Scale Up, Scale Out, Scale Across and beyond
- Liquid Cooling provides important design and reliability benefits

XPO broadens the architectural choices available to customers, enabling higher density while preserving front-panel pluggability.

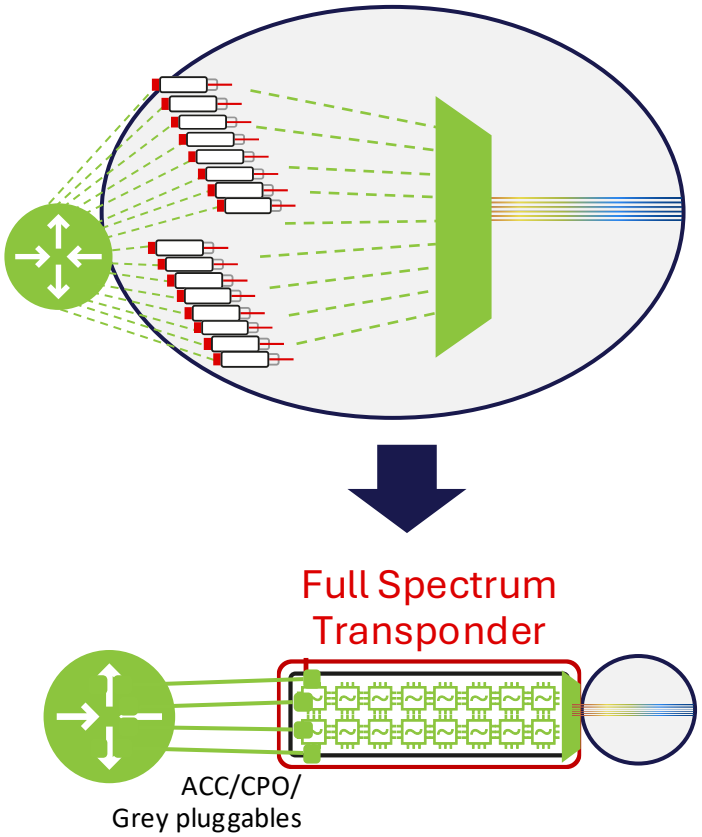
XPO
OPTICS INNOVATION




8 x XPO or 102.4T capacity

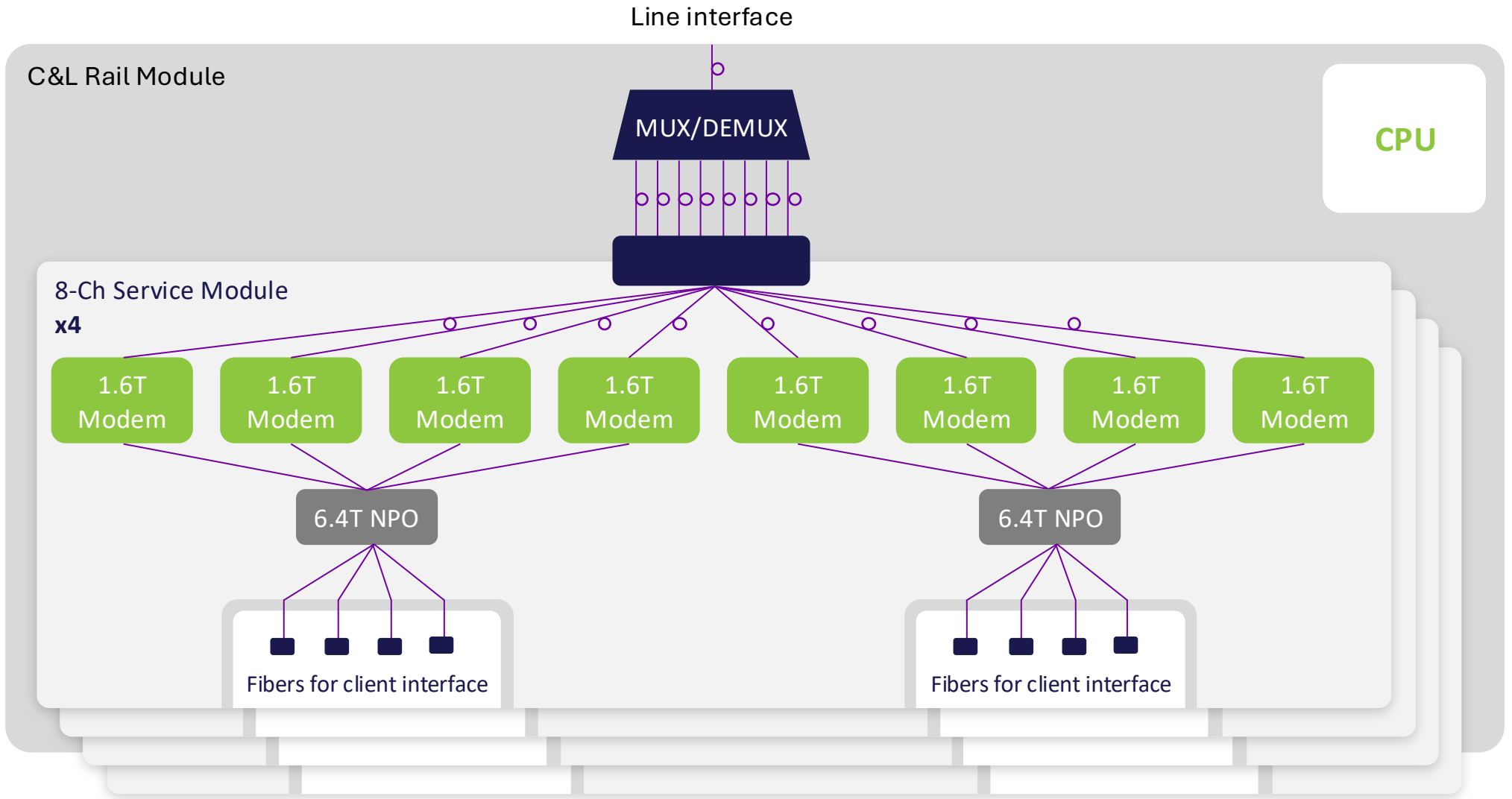
# At scale, the fiber becomes the unit of deployment

- Today: wavelength-by-wavelength deployment
  - 128 x 400G x 2 pluggables per fiber pair
  - 64 x 800G x 2 pluggables per fiber pair
- Scaling this model for 10s-100s fiber deployments becomes operationally complex.
- **Full spectrum transponder**
  - Lights the entire fiber at once
  - Returns to grey optics and external transponders
  - Reduces reliance on DWDM expertise
  - Enables cookie-cutter deployments
  - Can be higher performance without pluggable power constraints




Optimizing for deployment velocity increasingly favors full-fiber architectures

# What a 51.2T C&L full-spectrum transponder could look like



# Different constraints drive different optical architectures

| Power Efficiency | Spectral Efficiency | Deployment Velocity | Thermal/density limits |
|---|---|---|---|
| ↓ | ↓ | ↓ | ↓ |
| Coherent Pluggables | Performance Transponders | Full-Fiber Systems | New optical form factors |









The future of optical networking is not one architecture—but the coexistence of many.

# Call to Action

- Problem to solve
  - While multiple MSAs exist for pluggable form factors, it is important to standardize on mini quick disconnect connector that allows for standardized direct-to-plug liquid cooling interface
- OCP contribution:
  - Ciena and Staubli are contributing Mini Quick Disconnect (Mini-QD) Design Specification to the OCP Cold Plate Sub-project
  - Abstract has been accepted and contribution in progress
  - Contact bmohajer@ciena.com if you want to be involved
- Link to contribution DB/OCP Marketplace:

  https://www.opencompute.org/community/cold-plate

# Summary

- The future of optical networking is not one architecture—but the coexistence of many.
- Users community is encouraged to share requirements via active participation in public forum to promote standardization at a faster pace.
- Suppliers/vendors continue to work together via standards and MSAs to support the diverging range of coherent optics form factors and use cases.


Thank
You!
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October 12-15, 2026
San Jose, California